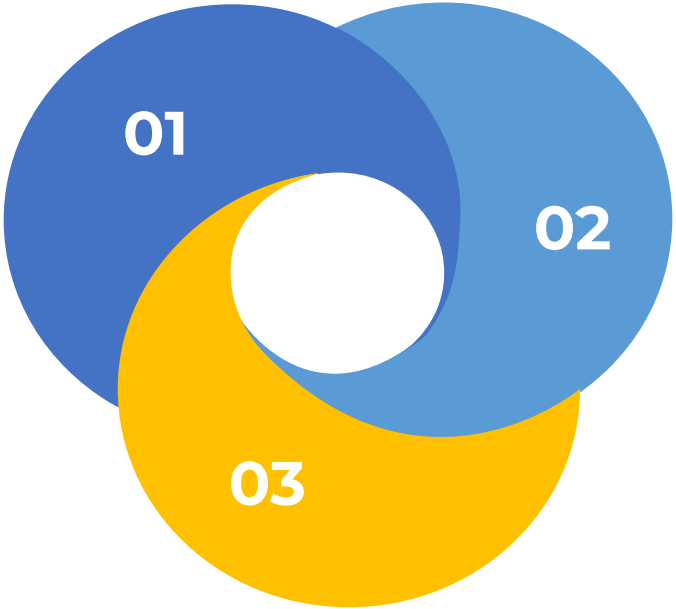
Statistical Analysis: Considerations

Significant considerations to conducting effective statistical analysis

Introduction

Statistical analysis is a cornerstone of research

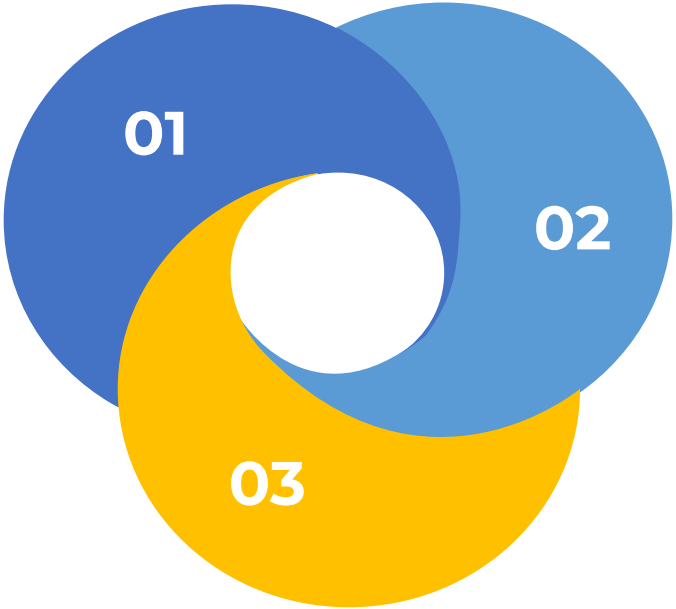


It provides insights into complex data

Supports evidence-based conclusions

Clear Research Question or Hypothesis

Start with a specific and clear question or hypothesis

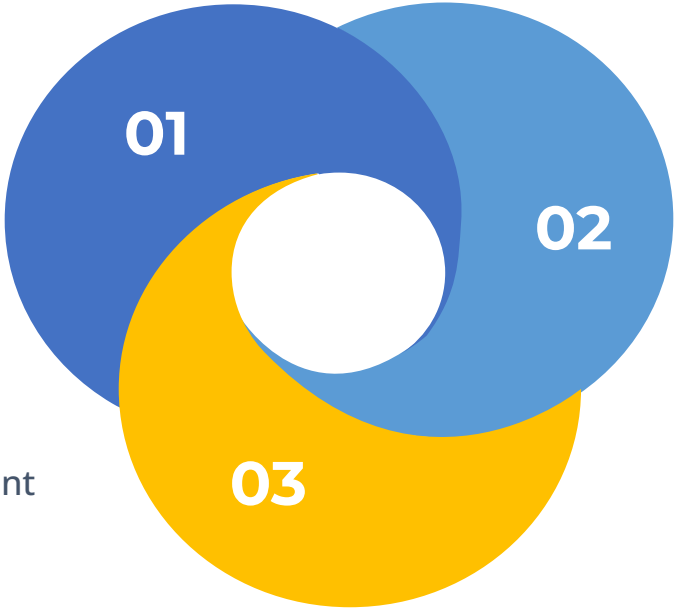


Helps in choosing the right tools for analysis

Sets the destination before starting the journey

Appropriate Data Collection

Collect representative data that reflects the population of interest

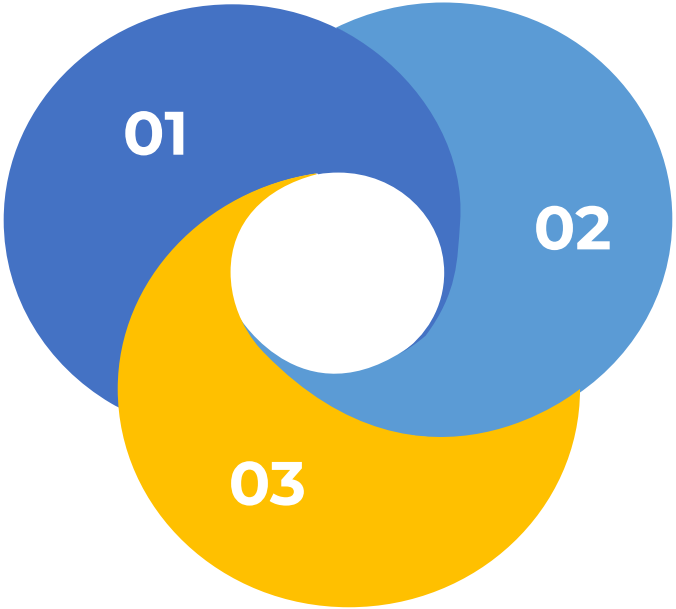


Follow a consistent process

Use accurate measurement tools

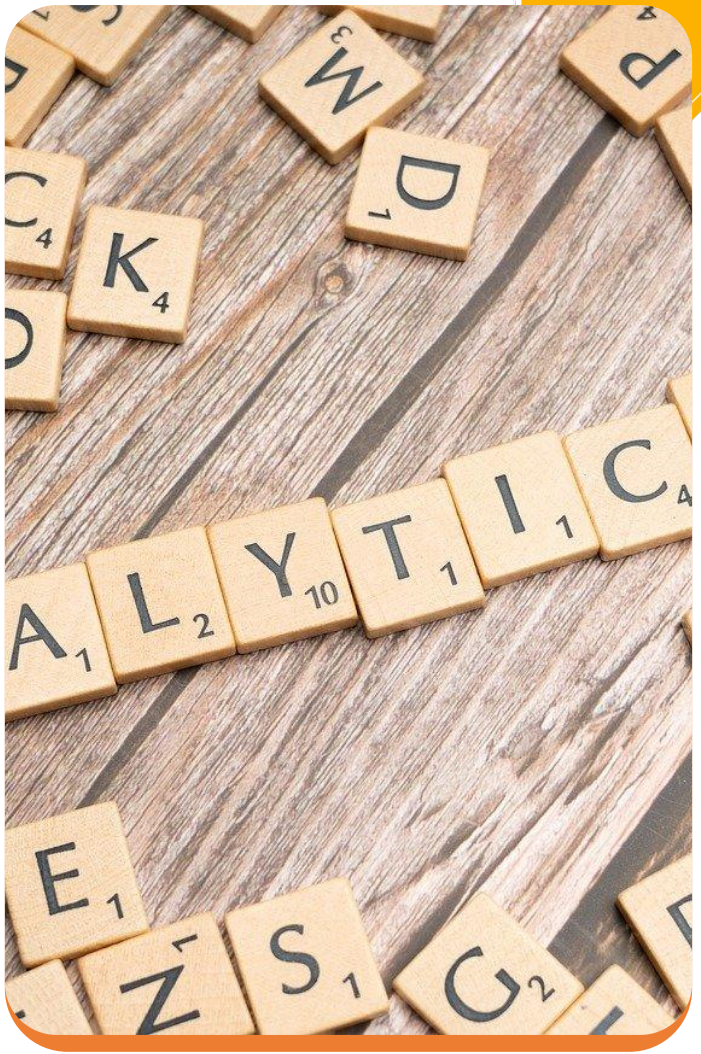
Data Quality

Ensure clean and reliable data



Identify missing pieces, errors, and oddities

Handle data problems thoughtfully



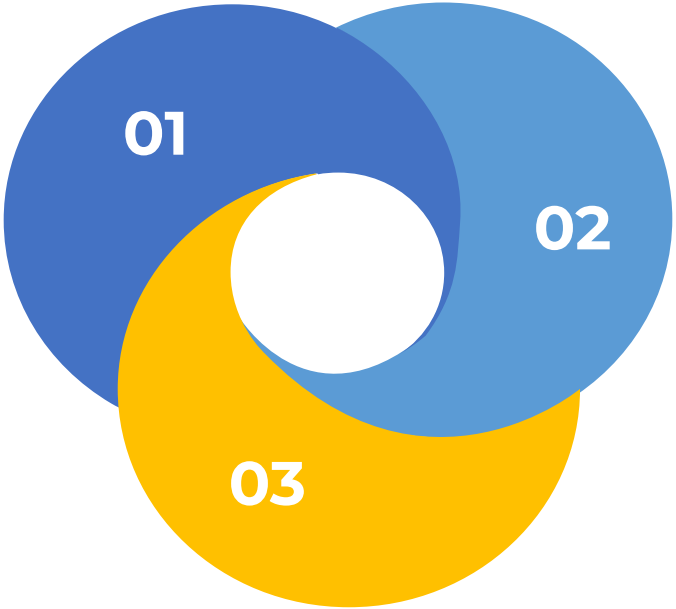
Selection of Statistical Methods

01 Consider the specific situation and characteristics of the data

02 Choose appropriate tools based on the data and research objectives

Understanding Assumptions

Know the assumptions underlying the chosen analysis methods



Check if the data matches the required assumptions

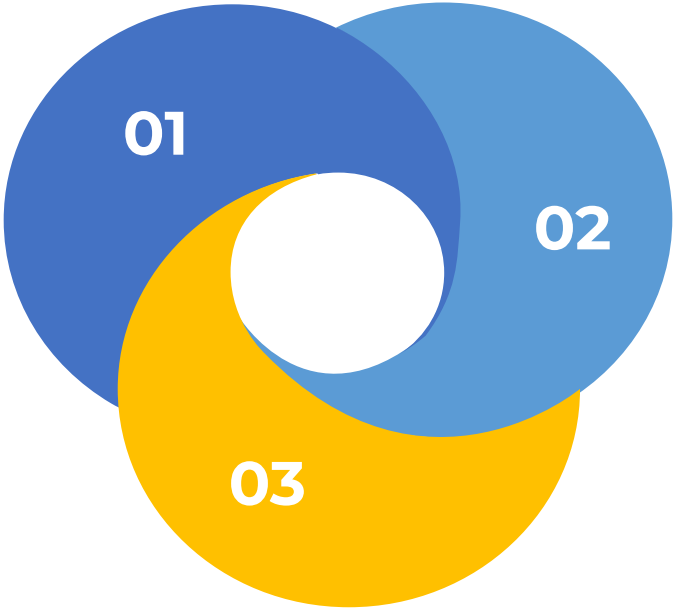
Avoid inaccurate results due to mismatched assumptions

Comprehensive Data Analysis

- Thoroughly examine the data before complex analyses
- Identify patterns, relationships, and potential issues
- Gain a holistic understanding of the data

Correct Interpretation of Results

Understand the meaning of analysis results

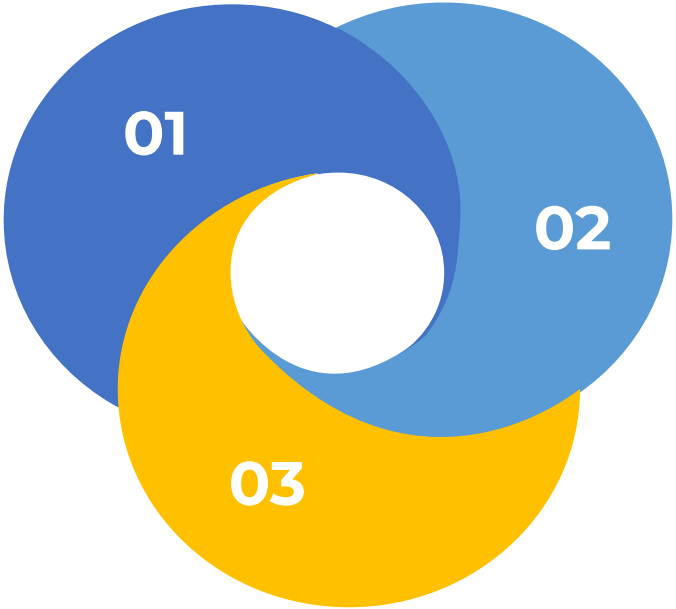


Relate them to the original question or hypothesis

Avoid premature conclusions

Reporting and Transparency

Clearly communicate the analysis process



Share findings, limitations, and challenges

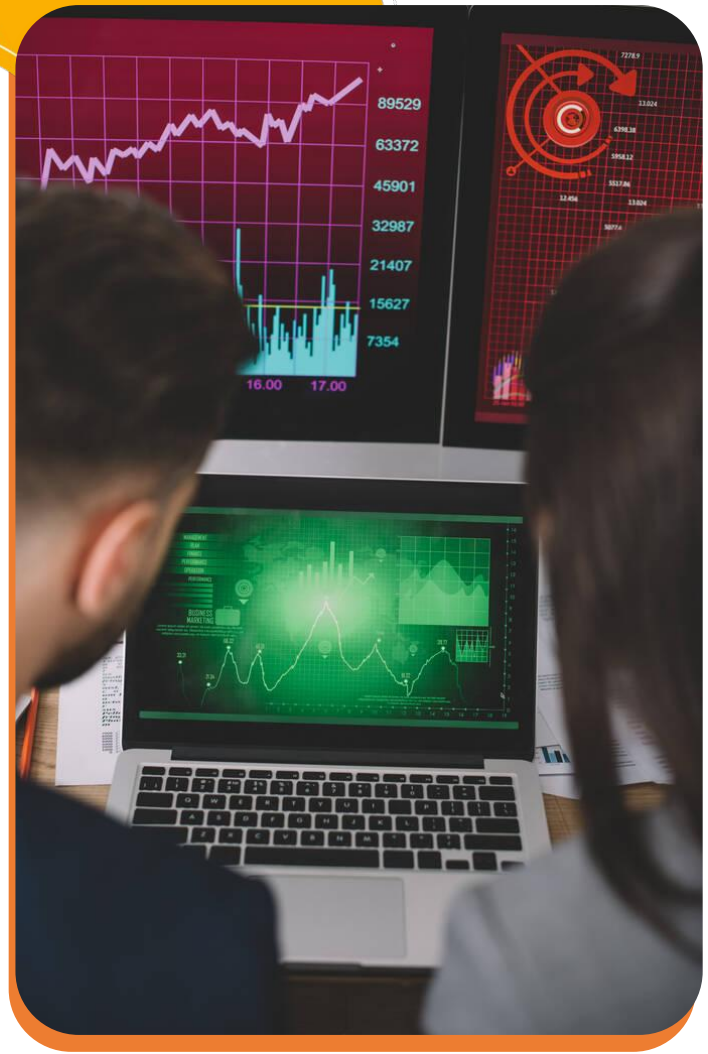
Promote trust and learning

Conclusion

01 Acknowledging and addressing limitations enhances research credibility

02 Effective statistical analysis is critical for deriving reliable insights

03 Requires careful planning, execution, and interpretation



Thank you for your time and attention 🙂